

Am25LS2536

Eight-Bit Decoder With Control Storage

DISTINCTIVE CHARACTERISTICS

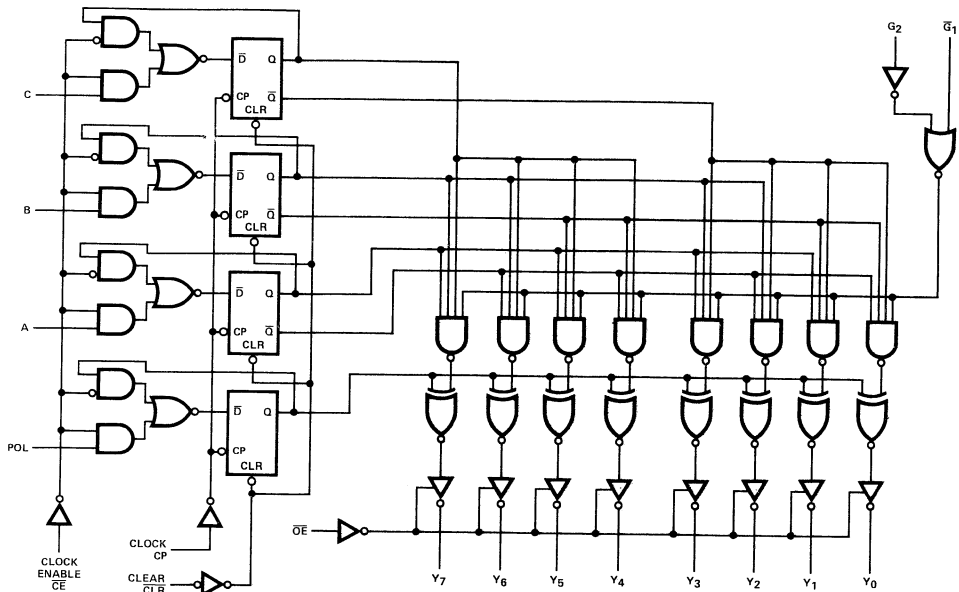
- 8-bit decoder/demultiplexer with control storage
- 3-state outputs
- Common clock enable
- Common clear
- Polarity control
- Advanced Low Power Schottky Process
- 100% product assurance screening to MIL-STD-883 requirements

FUNCTIONAL DESCRIPTION

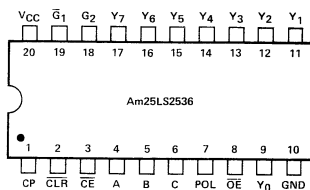
The Am25LS2536 is an eight-bit decoder with control storage. It provides a conventional 8-bit decoder function with two enable inputs which may also be used for data input. This can be used to implement a demultiplexer function. In addition, the exclusive "OR" gate allows for polarity control of the selected output. The 3-state outputs are enabled by a LOW on the ($\overline{OE}$) output enable.

The three control bits representing the output selection and the single bit polarity control are stored in "D" type flip-flops. These flip-flops have both Clear, Clock, and Clock Enable functions provided. The G_1 and G_2 input provide either polarity for input control or data.

LOGIC DIAGRAM
8-Bit Decoder/Demultiplexer with Control Storage

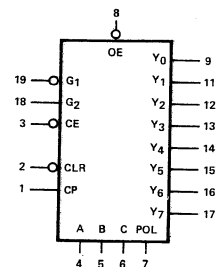


CONNECTION DIAGRAM



Note: Pin 1 is marked for orientation.

LOGIC SYMBOL



$V_{CC} = 20$
 $GND = 10$

ELECTRICAL CHARACTERISTICS

The Following Conditions Apply Unless Otherwise Specified:

COM'L $T_A = 0^\circ\text{C to } +70^\circ\text{C}$ $V_{CC} = 5.0\text{ V} \pm 5\%$ MIN. = 4.75 V MAX. = 5.25 V
 MIL $T_A = -55^\circ\text{C to } +125^\circ\text{C}$ $V_{CC} = 5.0\text{ V} \pm 10\%$ MIN. = 4.50 V MAX. = 5.50 V

DC CHARACTERISTICS OVER OPERATING RANGE

Parameters	Description	Test Conditions (Note 1)		Min.	Typ. (Note 2)	Max.	Units
V_{OH}	Output HIGH Voltage	$V_{CC} = \text{MIN.}$ $V_{IN} = V_{IH} \text{ or } V_{IL}$	$I_{OH} = -2.6\text{ mA, COM'L}$	2.4	3.2		Volts
			$I_{OH} = -1.0\text{ mA, MIL}$	2.4	3.4		
V_{OL}	Output LOW Voltage	$V_{CC} = \text{MIN.}$ $V_{IN} = V_{IH} \text{ or } V_{IL}$	$I_{OL} = 24\text{ mA, COM'L}$		0.4	0.5	Volts
			$I_{OL} = 12\text{ mA, MIL}$		0.35	0.4	
V_{IH}	Input HIGH Level	Guaranteed input logical HIGH voltage for all inputs		2.0			Volts
V_{IL}	Input LOW Level	Guaranteed input logical LOW voltage for all inputs	MIL			0.7	Volts
			COM'L			0.8	
V_I	Input Clamp Voltage	$V_{CC} = \text{MIN.}, I_{IN} = -18\text{ mA}$				-1.5	Volts
I_{IL}	Input LOW Current	$V_{CC} = \text{MAX.}, V_{IN} = 0.4\text{ V}$				-0.4	mA
I_{IH}	Input HIGH Current	$V_{CC} = \text{MAX.}, V_{IN} = 2.7\text{ V}$				20	μA
I_I	Input HIGH Current	$V_{CC} = \text{MAX.}, V_{IN} = 7.0\text{ V}$				0.1	mA
I_O	Off-State (High-Impedance) Output Current	$V_{CC} = \text{MAX.}$	$V_O = 0.4\text{ V}$			-20	μA
			$V_O = 2.4\text{ V}$			20	
I_{SC}	Output Short Circuit Current (Note 3)	$V_{CC} = \text{MAX.}$		-15		-85	mA
I_{CC}	Power Supply Current (Note 4)	$V_{CC} = \text{MAX.}$			37	56	mA

- Notes: 1. For conditions shown as MIN. or MAX., use the appropriate value specified under Electrical Characteristics for the applicable device type.
 2. Typical limits are at $V_{CC} = 5.0\text{ V}$, 25°C ambient and maximum loading.
 3. Not more than one output should be shorted at a time. Duration of the short circuit test should not exceed one second.
 4. Test Conditions: $A = B = C = \overline{G_1} = G_2 = \overline{OE} = \overline{CE} = \text{GND}$; $\text{CLK} = \text{CLR} = \text{POL} = 4.5\text{ V}$.

MAXIMUM RATINGS (Above which the useful life may be impaired)

Storage Temperature	$-65^\circ\text{C to } +150^\circ\text{C}$
Temperature (Ambient) Under Bias	$-55^\circ\text{C to } +125^\circ\text{C}$
Supply Voltage to Ground Potential Continuous	$-0.5\text{ V to } +7.0\text{ V}$
DC Voltage Applied to Outputs for High Output State	$-0.5\text{ V to } +V_{CC} \text{ max.}$
DC Input Voltage	$-0.5\text{ V to } +7.0\text{ V}$
DC Output Current, Into Outputs	30mA
DC Input Current	$-30\text{ mA to } +5.0\text{ mA}$

SWITCHING CHARACTERISTICS(T_A = +25°C, V_{CC} = 5.0V)

Parameters	Description		Min.	Typ.	Max.	Units	Test Conditions
t _{PLH}	G ₁ to Y ₀ – Y ₇			17	25	ns	C _L = 45pF R _L = 667Ω
t _{PLH}				23	34		
t _{PLH}	G ₂ to Y ₀ – Y ₇			20	30	ns	
t _{PHL}				26	39		
t _{PLH}	CP to Y ₀ – Y ₇			24	36	ns	
t _{PHL}				30	45		
t _{PLH}	CLR to Y ₀ – Y ₇			24	36	ns	
t _{PHL}				31	46		
t _s	Clock Enable to CP		25			ns	
t _h			0				
t _s	A, B, C, POL to CP		15			ns	
t _h			0				
t _{HZ}	OE to Y ₀ – Y ₇			9	14	ns	C _L = 45pF R _L = 667Ω
t _{LZ}				11	17		
t _{ZH}	OE to Y ₀ – Y ₇			15	22	ns	C _L = 45pF R _L = 667Ω
t _{ZL}				16	24		
t _s	Set-up Time, Clear Recovery to CP		20			ns	
t _{pw}	Pulse Width	Clock	15			ns	
		Clear	15				

**SWITCHING CHARACTERISTICS
OVER OPERATING RANGE***

Parameters			Description		Am26LS COM'L		Am25LS MIL		Units	Test Conditions
					T _A = 0°C to +70°C V _{CC} = 5.0V ±5%		T _A = -55°C to +125°C V _{CC} = 5.0V ±10%			
			Min.	Max.	Min.	Max.				
t _{PLH}	G ₁ to Y ₀ - Y ₇		29		31	ns	C _L = 45pF R _L = 667Ω			
t _{PHL}			39		42					
t _{PLH}	G ₂ to Y ₀ - Y ₇		34		37	ns				
t _{PHL}			44		48					
t _{PLH}	CP to Y ₀ - Y ₇		40		42	ns				
t _{PHL}			51		55					
t _{PLH}	CLR to Y ₀ - Y ₇		47		54	ns				
t _{PHL}			58		66					
t _s	Clock Enable to CP	27		30		ns				
t _h		0		0						
t _s	A, B, C, POL to CP	17		20		ns				
t _h		0		0						
t _{HZ}	OE to Y ₀ - Y ₇		17		18	ns	C _L = 5.0pF R _L = 667Ω			
t _{LZ}			27		34					
t _{ZH}	OE to Y ₀ - Y ₇		25		27	ns	C _L = 5.0pF R _L = 667Ω			
t _{ZL}			28		30					
t _s	Set-up Time, Clear Recovery to CP		23		25	ns				
t _{pw}	Pulse Width	Clock	17		20	ns				
		Clear	15		15					

*AC performance over the operating temperature range is guaranteed by testing defined in Group A, Subgroup 9.

FUNCTION TABLE

	Inputs									Internal Registers				Three-State Outputs								
Mode	C	B	A	POL	$\overline{CE}$	$\overline{CLR}$	G*	$\overline{OE}$	CP	Q _C	Q _B	Q _A	Q _{POL}	Y ₀	Y ₁	Y ₂	Y ₃	Y ₄	Y ₅	Y ₆	Y ₇	
Clear	X	X	X	X	X	L	L	L	X	L	L	L	L	H	H	H	H	H	H	H	H	H
	X	X	X	X	X	L	H	L	X	L	L	L	L	L	H	H	H	H	H	H	H	H
Hold	X	X	X	X	H	H	NC	L	↑	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
Select	L	L	L	H	L	H	H	L	↑	L	L	L	H	H	L	L	L	L	L	L	L	L
	L	L	H	H	L	H	H	L	↑	L	L	H	H	L	H	L	L	L	L	L	L	L
	L	H	L	H	L	H	H	L	↑	L	H	L	H	L	L	H	L	L	L	L	L	L
	L	H	H	H	L	H	H	L	↑	L	H	H	H	L	L	L	H	L	L	L	L	L
	H	L	L	H	L	H	H	L	↑	H	L	L	H	L	L	L	L	H	L	L	L	L
	H	L	H	H	L	H	H	L	↑	H	L	H	H	L	L	L	L	L	H	L	L	L
	H	H	L	H	L	H	H	L	↑	H	H	L	H	L	L	L	L	L	L	H	L	L
	H	H	H	H	L	H	H	L	↑	H	H	H	H	L	L	L	L	L	L	L	H	H
	L	L	L	L	L	H	H	L	↑	L	L	L	L	L	L	H	H	H	H	H	H	H
	L	L	H	L	L	H	H	L	↑	L	L	H	L	H	L	H	H	H	H	H	H	H
	L	H	L	L	L	H	H	L	↑	L	H	L	L	H	H	L	H	H	H	H	H	H
	L	H	H	L	L	H	H	L	↑	L	H	H	L	H	H	L	H	H	H	H	H	H
	H	L	L	L	L	H	H	L	↑	H	L	L	L	L	H	H	H	L	H	H	H	H
	H	L	H	L	L	H	H	L	↑	H	L	H	L	L	H	H	H	H	H	L	H	H
	H	H	L	L	L	H	H	L	↑	H	H	L	L	L	H	H	H	H	H	L	H	L
	H	H	H	L	L	H	H	L	↑	H	H	H	L	L	H	H	H	H	H	H	L	L
	X	X	X	H	L	H	L	L	↑	X	X	X	H	L	L	L	L	L	L	L	L	L
X	X	X	L	L	H	L	L	↑	X	X	X	L	H	H	H	H	H	H	H	H	H	
Output Disable	X	X	X	X	X	X	X	H	X	NC	NC	NC	NC	Z	Z	Z	Z	Z	Z	Z	Z	Z

*

$\overline{G_1}$	$\overline{G_2}$	G
L	L	L
L	H	H
H	L	L
H	H	L

NC = No Change X = Don't Care Z = High-Impedance ↑ = Low-to-High Transition

DEFINITION OF TERMS

$\overline{CLR}$ CLEAR – When the CLEAR input is LOW, the control register outputs (Q_A, Q_B, Q_C, Q_{POL}) are set LOW regardless of any other inputs.

CP CLOCK – Enters data into the control register on the LOW-to-HIGH transition.

$\overline{CE}$ CLOCK ENABLE – Allows data to enter the control register when $\overline{CE}$ is LOW. When $\overline{CE}$ is HIGH, the Q_i outputs do not change state, regardless of data or clock input transitions.

A, B, C Inputs to the control register which are entered on the LOW-to-HIGH clock transition if $\overline{CE}$ is LOW.

POL Input to the control register bit used for determining the polarity of the selected output.

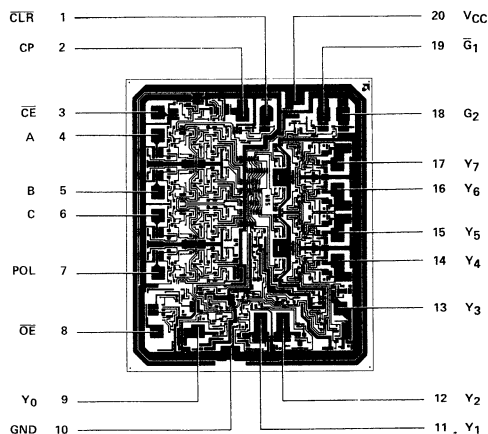
$\overline{G_1}$ Active LOW part of the expression $G = G_1G_2$ [or $G = (\overline{G_1})G_2$] where G is either data input for the selected Y_n or is used as an input enable.

G₂ Active HIGH part of the expression $G = G_1G_2$.

Y_n The three-state outputs. When active ($\overline{OE} = \text{LOW}$), one of eight outputs is selected by the code stored in the control register, with the polarity of all eight determined by the bit stored in the POL flip-flop of the control register. The selected output can further be controlled by G according to the expression $Y_{\text{SELECTED}} = G \oplus Q_{\text{POL}}$.

$\overline{OE}$ OUTPUT ENABLE. When $\overline{OE}$ is HIGH the Y_n outputs are in the high impedance state; when $\overline{OE}$ is LOW the Y_n's are in their active state as determined by the other control logic. The $\overline{OE}$ input affects the Y_n output buffers only and has no effect on the control register or any other logic.

Metallization and Pad Layout



DIE SIZE 0.084" X 0.099"